nature physics

Article

https://doi.org/10.1038/s41567-026-03466-2

Simultaneous high-fidelity single-qubit gates in a spin qubit array

Received: 4 November 2025

Accepted: 26 August 2026

Published online: 15 October 2026

Check for updates

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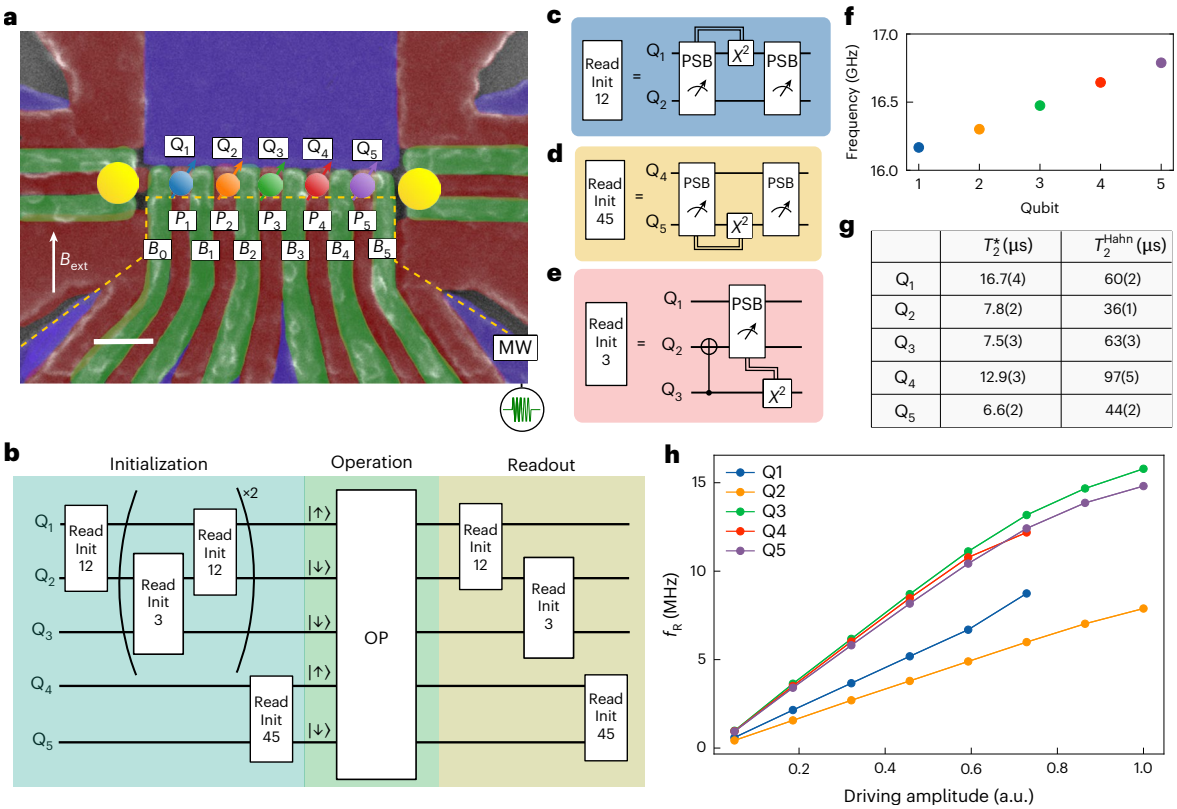
Silicon spin qubits offer a promising path to scalable quantum computing due to their compatibility with industrial semiconductor manufacturing. A key requirement for scaling quantum processors is the ability to perform high-fidelity operations in parallel across many qubits. However, simultaneous control remains a major challenge in silicon devices, as fidelities typically degrade under parallel operation. Here we identify microwave-drive-induced a.c.-Stark shifts as the dominant source of this degradation in a five-qubit silicon spin array. We address this by introducing a scalable mitigation protocol based solely on pairwise phase calibrations. Using tailored control pulses on a shared control line, we achieve primitive X and Y gate fidelities well above 99.99% for each qubit individually, with some approaching 99.999%. Furthermore, these fidelities are preserved above 99.99% during simultaneous operation of up to three qubits. During parallel five-qubit operation, fidelities remain above the widely targeted fault-tolerant threshold of 99.9%. By maintaining high-fidelity control during simultaneous operation, shared qubit-control lines can support the development of large-scale semiconductor quantum processors.

Quantum computing holds the potential to revolutionize fields from cryptography to materials discovery, but building scalable, fault-tolerant quantum processors remains a central challenge. Spin qubits in solid-state devices[1,2] are a leading platform candidate, with a compact footprint and compatibility with industrial semiconductor fabrication[3–6], offering a route towards scaling to the millions of qubits envisioned for fault-tolerant quantum computing architectures[7].

For fault-tolerant quantum computing, gate fidelities must exceed the threshold set by quantum error correction protocols, such as the surface code[8]. While the requirement depends on the noise model, fidelities above 99.9% are widely targeted for practical fault-tolerant operation[9,10]. In scalable quantum processors, control must extend beyond sequential single-qubit gates to the simultaneous operation of multiple qubits, because idle qubits dephase much faster than driven ones. In our silicon spin qubits, for example, driven coherence times are about $T_2^R \approx 100$ s, while idle dephasing times are only $T_2^* \approx 10$ s. This makes sequential operations impractical at scale because minimizing qubit idle times requires simultaneous control across many qubits. However, parallel qubit control typically degrades fidelities because of crosstalk. Such degradation has been observed not only in spin qubits[11–13] but also in superconducting systems, where mitigation strategies have been developed[14–17]. In spin qubits, the problem is particularly severe: fidelities were recently reported to drop considerably from 99.95% to 99.5% under parallel operation of only two qubits in silicon/silicon-germanium devices, the same platform used in this work[12]. The physical origin of this degradation has remained unclear[11–13].

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https://doi.org/10.1038/s41567-026-03466-2



	T_2^* (μs)	T_2^{Hahn} (μs)
Q_1	16.7(4)	60(2)
Q_2	7.8(2)	36(1)
Q_3	7.5(3)	63(3)
Q_4	12.9(3)	97(5)
Q_5	6.6(2)	44(2)

Fig. 1 | Five-qubit device. a, False-coloured scanning electron micrograph of a cofabricated device identical to the one used in this experiment (scale bar, 100 nm). An array of five quantum dots is accumulated beneath plunger gates $P_{1\text{-}5}$, hosting the five spin qubits. Two sensor quantum dots (yellow) are accumulated on both ends of the array, allowing single-shot readout. Microwave signals are applied to the MW screening gate (outlined by the dashed line) to perform single-qubit operations via EDSR. A cobalt micromagnet is fabricated on top of the device (not shown) to enable qubit addressability and EDSR operations within a gradient field. **b**, Device operation scheme. The five qubits are initialized in the $|\uparrow\downarrow\downarrow\uparrow\downarrow\rangle$ state, manipulated during the operation stage using control pulses, and then read out. **c**,**d**, Initialization (Init) and measurement of qubit pairs $Q_{1,2}$ (**c**) and $Q_{4,5}$ (**d**) using PSB readout and a feedback X^2 gate conditioned on the PSB readout result. All PSB integration times are set to 10 μs. **e**, Initialization and measurement of qubit Q_3 using QND readout and a conditional X^2 gate. **f**, Qubit resonance frequencies at an external field of $B = 0.42$ T. The micromagnet gradient induces an approximate 150-MHz separation between all adjacent qubits, enabling individual qubit addressability with MW pulses. **g**, Qubit phase coherence time T_2^* and Hahn echo sequence coherence time T_2^{Hahn} for all qubits. The total integration time is a few minutes for all measurements. **h**, The driving amplitude dependence of the qubit Rabi frequencies f_R shows a linear relationship for all qubits up to $f_R \approx 8$ MHz, after which some qubits begin to exhibit nonlinear behaviour.

Addressing it is therefore essential for implementing large-scale error correction codes that require parallel gate execution.

We identify microwave-drive-induced a.c.-Stark shifts as the dominant source of fidelity degradation during parallel operation and introduce a compact calibration protocol that compensates these shifts using only pairwise measurements. This approach avoids the exponential overhead of full crosstalk characterization and enables high-fidelity parallel control of five qubits through a shared qubit-control line. Tailored control pulses allow individual qubits to reach fidelities well above 99.99%, with some approaching 99.999%, as measured by randomized benchmarking (RB). Fidelities above 99.99% are also sustained during simultaneous operation of up to three qubits, while full five-qubit parallel operation maintains fidelities at the practical fault-tolerant threshold of 99.9% or higher. This performance exceeds previously reported fidelities in silicon spin qubits and establishes a benchmark for scalable, high-fidelity parallel control in semiconductor quantum processors. The use of a shared control line further reduces wiring complexity, supporting continued scaling towards larger qubit arrays in silicon.

Five-qubit device and operation

Figure 1a shows the false-coloured scanning electron micrograph of a cofabricated silicon five-qubit device identical to the one used in this experiment. The device is fabricated in an isotopically purified ^{28}Si/SiGe quantum well. Three overlapping aluminium metal gate layers, separated by native oxide, are deposited on the wafer to define and control an array of five quantum dots that host spin qubits. Two adjacent charge sensor quantum dots at the ends of the array are used for qubit readout[18]. A cobalt micromagnet (not shown) is deposited on top of the device to induce a gradient field for qubit addressability and operation[19,20]. A microwave signal (MW) is applied to the MW gate electrode (dashed line) to manipulate the spin states of all five qubits in the array via electric dipole spin resonance (EDSR)[21–23], thereby providing a shared control line for all qubits. Further details on the device are given in Methods.

Figure 1b shows the device operation scheme. We operate the device in the (3,1,1,3,1)-electron charge configuration for the five quantum dots. Pauli-spin blockade (PSB) is used to read out the parity of qubit pairs $Q_{1,2}$ and $Q_{4,5}$ by projecting qubit Q_2 (Q_5) onto qubit Q_1 (Q_4). The dots containing three electrons serve as the projection dots for the PSB readouts, which helps increase the readout windows[18]. We use feedback initialization[18,24] to initialize the qubit states. The circuits for qubit pairs $Q_{1,2}$ and $Q_{4,5}$ are shown in Fig. 1c,d. The qubit parity is first read out using PSB. After the PSB measurement, the qubits are pulsed back to the (3,1) operation point. An X^2 operation is applied to perform a π-flip on qubit Q_1 or Q_5 if an even parity state ($|\downarrow\downarrow\rangle$ or $|\uparrow\uparrow\rangle$) is measured, thereby flipping the system into an odd parity state ($|\uparrow\downarrow\rangle$ or $|\downarrow\uparrow\rangle$). No operation is applied if an odd parity state is measured. Another PSB

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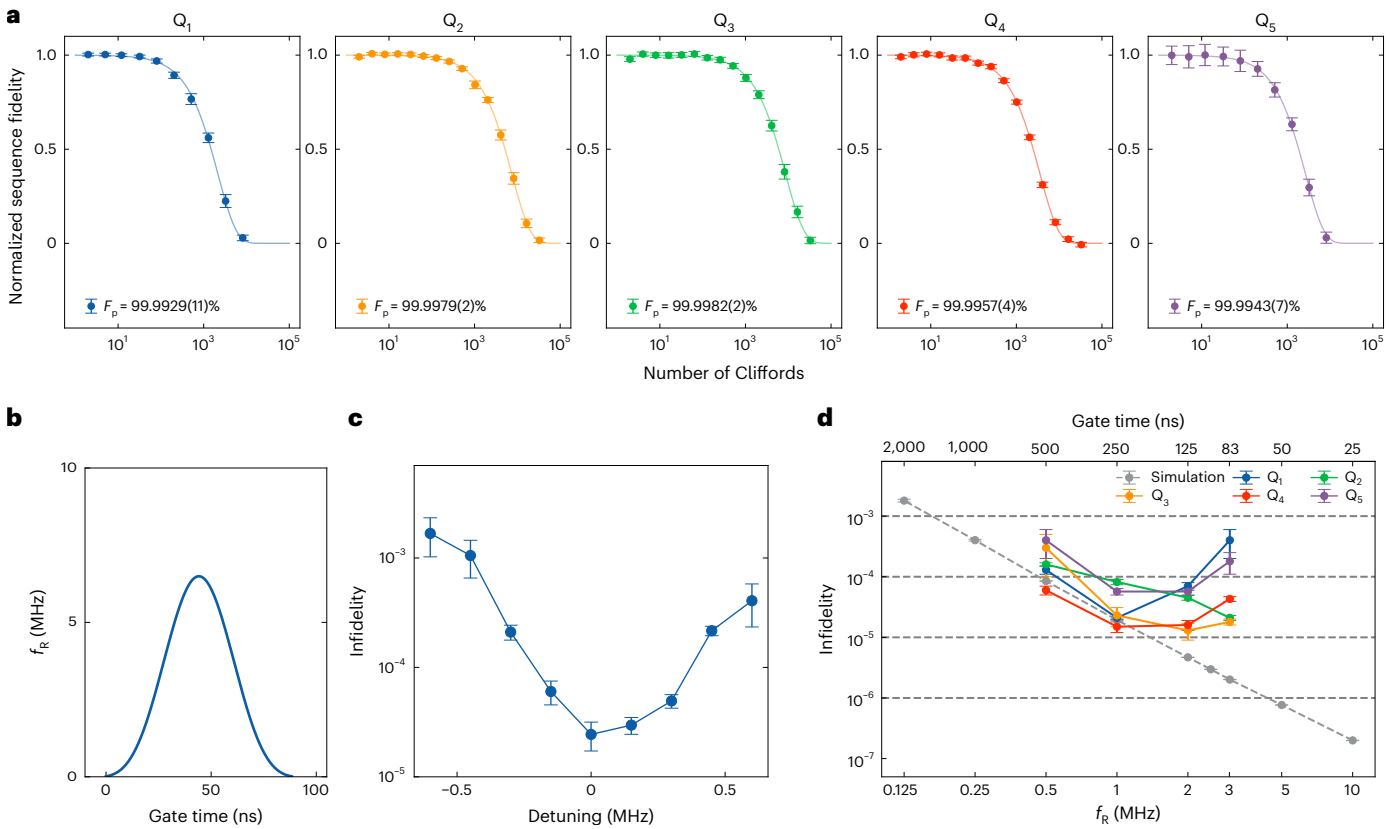


Fig. 2 | High-fidelity single-qubit gates. a, Single-qubit RB experiments with individually addressed qubits using Kaiser-window pulses. All five qubits reach primitive π/2 gate fidelities F_p above 99.99%. The gate times of the primitive gates for the five qubits are 125 ns, 83 ns, 83 ns, 83 ns and 125 ns, respectively. Data are averaged over 35 randomized sequences per Clifford length. Error bars show 95% confidence intervals; fidelity uncertainties are obtained from the fitting covariance matrix. Data points are normalized with fitted decay amplitude (Methods). **b**, Kaiser-window control pulse used in the RB experiments. **c**, RB fidelities of qubit Q_3 under MW frequency detuning. RB measurements show fidelities >99.99% over a range of ≈0.4 MHz. The slight skew towards positive detuning is attributed to MW-induced heating (Extended Data Fig. 7 and Supplementary Section 2). Points show the mean of four repetitions; uncertainties are obtained from the fitting covariance matrix. **d**, Single-qubit RB at varying operation speeds using Kaiser-window pulses. The operation speed is defined as the Rabi frequency f_R of a square pulse with the same area as the applied pulse. Simulations use noise extracted from independent measurements of qubits under idling conditions (Extended Data Fig. 4), which defines a lower bound set by the dephasing limit. At faster gate speeds, the deviation of the experimental data from the simulations results from MW-induced decoherence caused by the control pulses, not captured in the simulations. These effects become more pronounced at higher operation speeds, as faster gates require stronger MW drive. Data are averaged over 35 randomized sequences per Clifford length. Fidelity uncertainties are obtained from the fitting covariance matrix.

readout is performed following this feedback operation to project the state into the $S(4, 0)$ singlet state. The micromagnet gradient lifts the degeneracy between the spin states, allowing initialization of the $S(4, 0)$ singlet state into the $|\uparrow\downarrow\rangle$ state via an adiabatic ramp. We use the (3,1,1,3,1) charge configuration and the corresponding projections to maximize the readout window of both PSB readouts. Qubit Q_3 is measured by quantum non-demolition (QND) measurement between Q_2 and Q_3 (ref. 25), as shown in Fig. 1e. A CROT gate is applied with qubit Q_3 as the control qubit and Q_2 as the target. A PSB readout of the $Q_{1,2}$ pair is then performed, and if an even (odd) parity state is measured, we infer that qubit Q_3 is in the $|\uparrow\rangle$ ($|\downarrow\rangle$) state. Based on this readout, a conditional X^2 gate is applied to qubit Q_3, preparing it in the $|\downarrow\rangle$ state. After the QND readout, the $Q_{1,2}$ pair is re-initialized using the PSB feedback initialization circuit. The QND readout is repeated twice to increase the initialization fidelity of qubit Q_3.

The five-qubit state is initialized to $|\uparrow\downarrow\uparrow\downarrow\uparrow\rangle$ before the operation stage, during which single- and two-qubit operations are applied to control the quantum processor. Afterward, the parity of qubit pairs $Q_{1,2}$ and $Q_{4,5}$, along with the state of Q_3, is read out, yielding three bits of information per cycle. The resonance frequencies of the five qubits, shown in Fig. 1f, exhibit a Zeeman energy difference of $\Delta E_Z \approx 150$ MHz between neighbouring qubits, consistent with our micromagnet design. Coherence properties are characterized by an average dephasing time $T_2^* \approx 10.3$ s and an average Hahn echo decay time $T_2^{\mathrm{Hahn}} \approx 60$ s, as shown in Fig. 1g. Low residual exchange is crucial for high-fidelity single-qubit gates in multiqubit systems[26,27]. Under single-qubit operating conditions, we measure the residual exchange below $J < 16.6$ kHz for all neighbouring qubit pairs (Extended Data Fig. 1). For two-qubit gate operations, the exchange coupling is activated into the megahertz range by pulsing the voltages on the interdot barrier gates B1–B4 (Methods).

High-fidelity single-qubit gates

We first operate each qubit individually. Single-qubit RB experiments are performed on all five qubits using Kaiser-window control pulses, shown in Fig. 2b. The Kaiser-window pulse features an excitation spectrum engineered to suppress direct MW-drive-induced qubit-qubit crosstalk from spectral leakage[26], which is essential for spin qubit arrays operated with a shared control line. A key requirement for achieving high-fidelity control with tailored pulses such as the Kaiser window is to operate the qubits in a regime where the Rabi frequency scales linearly with the drive amplitude. To verify this, we measure the Rabi frequency f_R as a function of drive amplitude for all five qubits and confirm linearity up to $f_R \approx 8$ MHz, as shown in Fig. 1h. Beyond 9 MHz,

qubit Q_1 exhibits reduced driven coherence time T_2^R. Based on this observation and additional considerations from the tailored pulses, we choose a minimal gate time of $t_g = 83$ ns.

The Clifford gate set is composed solely of X and Y gates, each corresponding to a π/2 pulse, such that the fitted primitive gate fidelity matches the fidelity of the single-qubit π/2 gate (Methods). To minimize qubit idling, where they are more susceptible to dephasing noise, we replace the identity gate I in the gate set with $4 \times X$ or $0 \times X$ (skipping the gate completely). Both replacements result in similar fidelities, enhancing the robustness of the gate sequences against dephasing[11]. For the same reason, the idle time between primitive quantum gates is minimized to 2 ns, the hardware limit of our control equipment (Methods). The qubit frequency and the amplitude for the single-qubit gates are calibrated and fine-tuned before the experiment (Extended Data Fig. 2). Following these calibrations, we measure the RB decay for all five qubits, as shown in Fig. 2a. All qubits reach primitive single-qubit fidelities above 99.99%, with some approaching 99.999%. Interleaved RB[28] confirms that the X and Y gates perform similarly, both achieving fidelities well above 99.99% (Extended Data Fig. 3).

Next, we test the robustness of our high-fidelity control against frequency detuning by measuring the RB fidelity of qubit Q_3 as a function of MW–frequency offset, as shown in Fig. 2c. We find that fidelities remain above 99.99% across a detuning window of ≈0.4 MHz. Notably, this window is larger than the typical qubit frequency fluctuations in our devices (≈0.1 MHz; Extended Data Fig. 4), which are primarily caused by charge noise[20,29,30].

Gate fidelity depends on a trade-off between dephasing at low speeds and reduced driven coherence at high speeds. We evaluate this behaviour by performing single-qubit RB experiments at varying operation speeds (Fig. 2d). The resulting infidelity increases both for slow gates, owing to charge-noise-induced dephasing[31], and for fast gates, owing to additional MW-induced decoherence[20,32–34]. This trade-off is reflected in the qubit operation quality factors, which exhibit a maximum at the optimal driving speed (Extended Data Fig. 5). For qubits Q_1 to Q_4, the optimal gate time results in a fidelity close to 99.999%, while Q_5 reaches an optimal fidelity of ≈99.995%, consistent with its lower maximum spin-lock quality factor. We attribute the variations between qubits to differences in the confinement potential, probably arising from local fabrication, tuning conditions and material variability common in academic devices such as the one used here[23,35]. The proximity of qubits Q_1 and Q_5 to the charge sensors acting as reservoirs may also render them more susceptible to drive-induced decoherence, resulting in lower spin-lock quality factors.

To quantitatively assess these trends, we compare the experimental fidelities to numerical simulations based on dephasing noise measured under idling conditions (Extended Data Fig. 4). As expected, the simulations accurately reproduce the infidelity trends at slower gate speeds (500 ns to 250 ns), consistent with the expected infidelity scaling $1 - F \propto t_g^2$ for dephasing under quasi-static noise (see Methods for details). However, at faster gate speeds, the experimental fidelities are lower than predicted. This indicates that MW-induced noise from qubit operations, absent during idling and thus not included in the model, emerges at faster speeds and contributes to decoherence. The associated infidelity can be reproduced by modelling a moderate increase in the qubit noise level caused by the applied MW signals (Extended Data Fig. 4).

Simultaneous operation of two qubits

As a first demonstration of high-fidelity parallel control, we investigate simultaneous operation of two qubits. We benchmark the fidelity of this parallel control using a simultaneous RB (SRB) experiment on qubits Q_3 and Q_4 (ref. 14). In this experiment, independent randomized Clifford sequences are applied simultaneously to both qubits, and the resulting decays are fitted to extract individual gate fidelities under simultaneous operation (Methods). Simultaneous quantum gates are implemented by combining the single-qubit MW pulses for each qubit using a MW combiner, enabling operation of multiple qubits with a shared control line (see also Methods). Because of the a.c.-Stark effect, MW control pulses addressed to one qubit also expose neighbouring qubits on the shared control line to off-resonant fields, shifting their resonance frequencies and inducing z-phase errors (Extended Data Fig. 6). These crosstalk phase shifts are quadratic in the off-resonant drive amplitude, with the sign opposite for each qubit in the pair, as is characteristic of the a.c.-Stark effect. Simultaneous operations also modify the optimal MW amplitudes required for π/2 rotations, indicating residual driven-gate errors beyond the crosstalk phase shifts. To achieve high-fidelity simultaneous operations, both the residual rotation errors and z-phase errors must be calibrated and compensated.

To calibrate the crosstalk phase errors, we apply a Hahn-echo sequence to the target qubit while performing $4NX$ gates on another qubit (Fig. 3a,b) and measure the resulting phase accumulation per gate (Fig. 3c,d). The crosstalk phase accumulated on qubit Q_i when operating qubit Q_j is denoted as $\Delta\phi_{ij}$. For Q_3 and Q_4, the measured crosstalk phases are $\Delta\phi_{34} = 0.0379(2)$ rad and $\Delta\phi_{43} = -0.0026(1)$ rad. The discrepancy in phase shift magnitudes arises from the different power required to drive each qubit at the same operation speed (Extended Data Fig. 5): Q_3 requires less power than Q_4, resulting in a smaller off-resonant drive and, consequently, a smaller a.c.-Stark shift experienced by Q_4. These phases are compensated during the simultaneous pulse sequence by offsetting the MW phases during the experiment, as illustrated in Fig. 3e. While we apply a fixed phase compensation per pulse, this corrects the dominant crosstalk-induced phase shifts during simultaneous operation. Time-dependent phase modulation within the pulse envelope would enable direct compensation of amplitude-dependent phase shifts during the gate. Slower phase drifts, for example, due to microwave-induced heating (Extended Data Fig. 7), lead to additional phase accumulation on longer timescales[36], suggesting that dynamical phase compensation could further improve gate fidelity.

After measuring the crosstalk phase errors, we calibrate the operation pulse amplitudes, adjusting A_3 and A_4 for the two-qubit simultaneous operation (Fig. 3f). We apply 16 simultaneous X gates to the qubit pairs, followed by 1 (3) simultaneous X gate(s), which flip the qubits into $|+x, +x\rangle$ ($|-x, -x\rangle$) state to maximize sensitivity to over- or under-rotation errors. Because the probabilities P_{+X} and P_{-X} for the two sequences are equal at an optimal MW amplitude, we use an optimizer to minimize the summed probability difference $\sum_{Q_3,Q_4} |P_{+X} - P_{-X}|$ of both qubits to obtain the optimal pulse amplitudes. Figure 3f shows a two-dimensional (2D) map of the probability difference, along with an illustration of the optimization process.

To demonstrate the effect of our calibration scheme, we perform a two-qubit SRB experiment on qubits Q_3 and Q_4, both with and without applying the calibrations for the simultaneous operations (Fig. 3g). In the uncalibrated case, the individual single-qubit gate pulses are summed and applied to qubit Q_3 and Q_4 directly. Without calibration, Q4 still achieves a gate fidelity above 99.99%, while Q3 remains limited to 99.9% due to a sizable uncorrected phase accumulation $\Delta\phi_{34}$. After applying the calibrations, both qubits reach fidelities exceeding 99.99%.

Simultaneous operation of multiple qubits

We now extend our parallel-control experiments to multiple qubits in the array. The crosstalk phases are calibrated for multiqubit operations by measuring the phases accumulated on each qubit individually and summing these contributions linearly. To validate this approach, we measure the crosstalk phases accumulated on Q_3 when operating Q_2 or Q_4 individually (see Fig. 4a,b for the quantum circuit) and compare them with simultaneous pulses on both qubits (Fig. 4c). The measured phases as a function of control pulse amplitude are shown in Fig. 4d. Summing the fitted individual-phase results (dashed black line) agrees well with simultaneous pulse measurements, demonstrating

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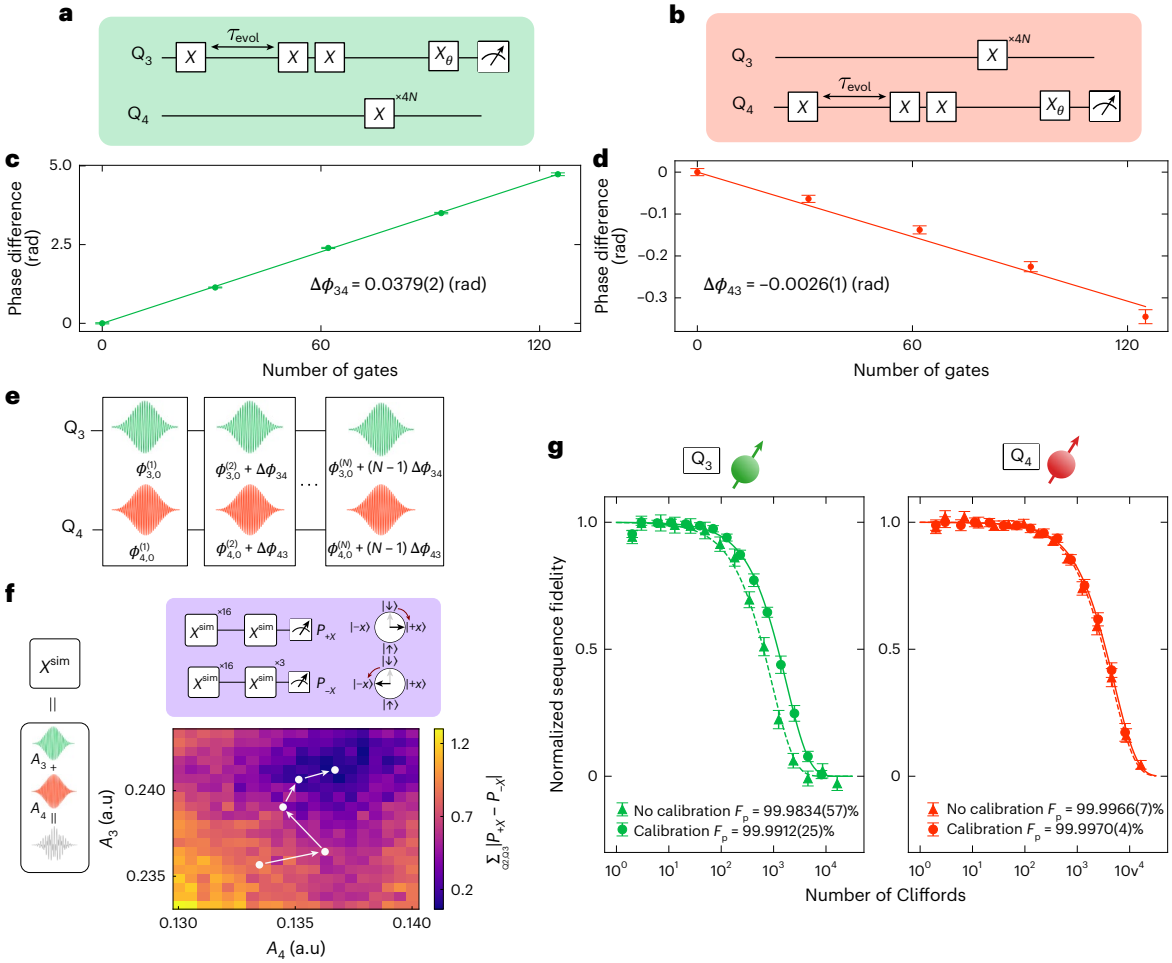


Fig. 3 | Crosstalk phase in simultaneous two-qubit operation. a,**b**, Sequence for measuring the crosstalk phase errors accumulated (**a**) on qubit Q_3 by Q_4 control pulse and (**b**) on qubit Q_4 by Q_3 control pulse. **c**,**d**, Measured crosstalk phase errors accumulated on qubit Q_3 when operating qubit Q_4 (**c**) and on qubit Q_4 when operating qubit Q_3 (**d**). Each point is a phase extracted from a cosine fit; error bars are obtained from the fitting covariance matrix. **e**, Compensation of crosstalk phase error during simultaneous operations. For each pulse, the MW phase in the following gates on qubit Q_3 (Q_4) is shifted by $\Delta\phi_{34}$ ($\Delta\phi_{43}$). $\phi_{3,0}^{(n)}$ and $\phi_{4,0}^{(n)}$ correspond to the uncompensated phase values, which are equal to 0 (π/2) if an X (Y) gate is applied. **f**, Calibration of the operation amplitudes A_3 and A_4 for simultaneous operations on Q_3 and Q_4. The simultaneous X gate is implemented by combining the individual qubit driving signals and applying them to the MW gate. A Nelder–Mead optimizer from SciPy[46] is used to minimize the probability difference and extract optimal amplitudes. The 2D map shows the probability difference between the two sequences, with the white trace indicating the optimization trajectory. Only a sparse subset of points (white circles) is sampled during optimization. **g**, SRB decays of qubit Q_3 and Q_4 without and with the phase and amplitude calibration. With calibration, Q_3 shows a substantial fidelity increase, while Q_4 remains mostly unaffected, due to a larger crosstalk phase error in Q_3. In this SRB scheme, the RB sequences differ for the two qubits. Data are averaged over 35 randomized sequences per Clifford length. Error bars show 95% confidence intervals; fidelity uncertainties are obtained from the fitting covariance matrix.

that crosstalk phases can be measured pairwise and summed linearly. This method enables precise, scalable compensation of phase crosstalk in multiqubit control sequences.

Figure 4e,f illustrates the calibration procedure for general multi-qubit simultaneous operations, demonstrated here for the three-qubit case with qubits Q_2, Q_3 and Q_4. The simultaneous π/2 gate duration is set to 250 ns to reduce the peak driving amplitude. Crosstalk phases $\Delta\phi_{ij}$ between each qubit pair are measured individually and compensated within the gate sequences by shifting the MW phase on each qubit Q_i by $\sum_{j\neq i}\Delta\phi_{ij}$, as illustrated in Fig. 4f. The simultaneous operation amplitudes are then calibrated using the same method described for the two-qubit case. Figure 4g illustrates the three-qubit optimization process, showing the driving amplitudes A_2, A_3 and A_4, and the resulting probability difference at each step. After approximately ten steps, the amplitudes converge, minimizing the probability difference. Using these calibrated phases and amplitudes, we perform a three-qubit SRB experiment and achieve primitive π/2 gate fidelities >99.99% for all three qubits (Fig. 4h).

Finally, we explore the limits of parallel control with our shared control line approach by operating all five qubits simultaneously and discuss the implications for larger arrays. To achieve this, we calibrate all crosstalk phases and control pulse amplitudes using the pairwise-calibration procedure described above. We also compare this pairwise-calibration approach with a full calibration, in which the phase accumulation on each qubit is measured while the other four are operated, and observe no difference in performance. This underscores that the pairwise approach enables scalable and resource-efficient parallel control across multiple qubits. Due to readout limitations (only three bits of information per cycle), we perform tomographic readout[18] on the qubit pairs $Q_{1,2}$ and $Q_{4,5}$ to reconstruct the individual qubit states (Fig. 5a; Methods). The complete five-qubit SRB sequence is illustrated in Fig. 5b, where each random Clifford sequence is repeated twice to perform the two required tomographic measurements (ZZ and ZI) needed to reconstruct the full five-qubit state.

We perform SRB experiments with gate durations of t_g = 250 ns and 500 ns (Fig. 5c). In both cases, overall fidelities remain similar but lower

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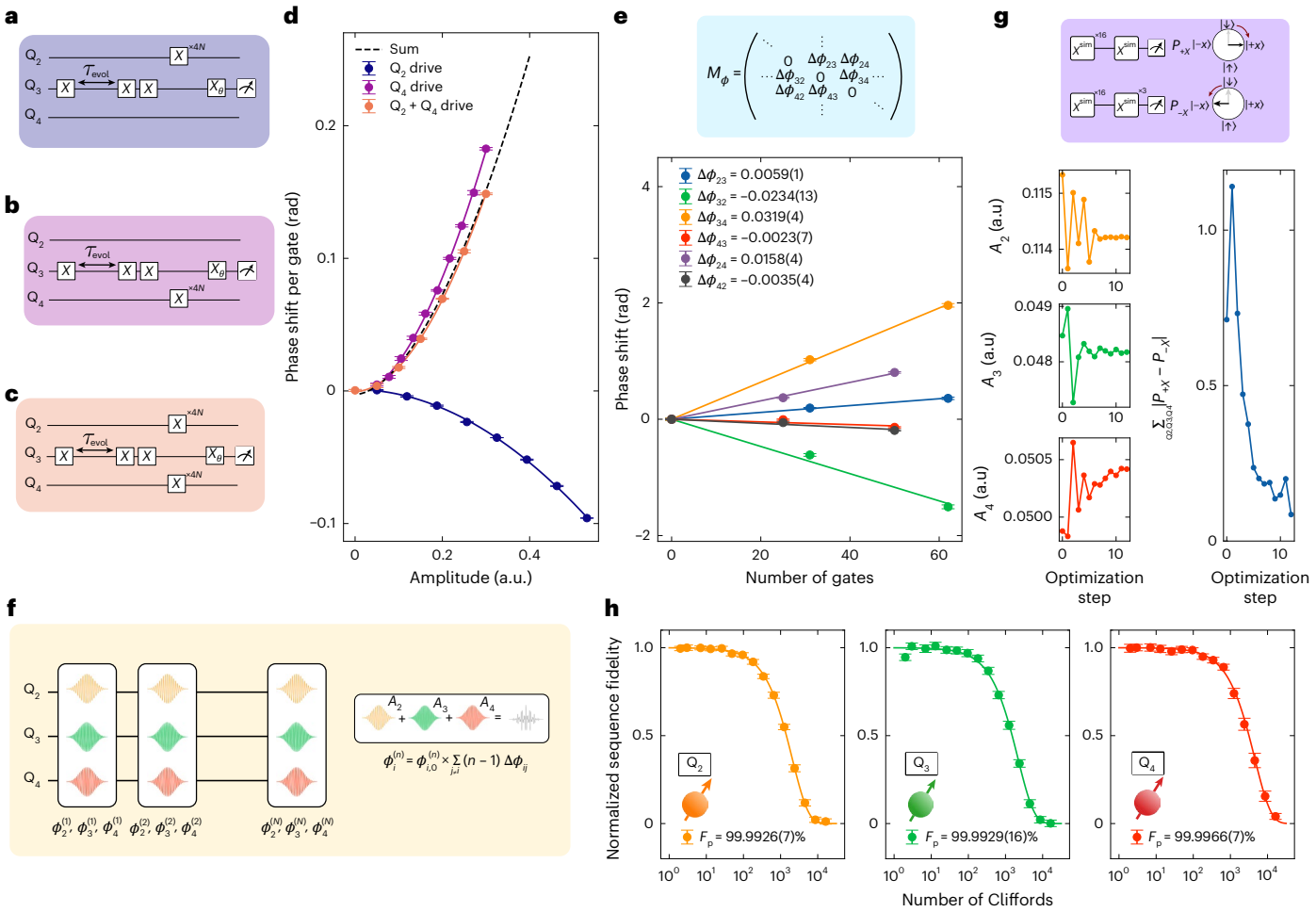


Fig. 4 | Generalized calibration of parallel multiqubit operations. **a**–**c**, Sequence for measuring crosstalk phase error accumulated on qubit Q_3 while operating (**a**) qubit Q_2, (**b**) qubit Q_4 and (**c**) both Q_2 and Q_4 simultaneously. **d**, Amplitude dependence of the crosstalk phases. We fit the phase accumulations with AV_{amp}^{α} and repeat the experiment for other qubit pairs. We find an average $\alpha = 1.91(5)$, in good agreement with the a.c.-Stark shift (Extended Data Fig. 6). The dashed black line shows the sum of the two fits in the individual operation cases. **e**, Crosstalk phase measured for all qubit pairs in a qubit $Q_{2,3,4}$ three-qubit simultaneous drive. In **d** and **e**, each point is a phase extracted from a cosine fit; error bars are obtained from the fitting covariance matrix. **f**, Illustration of the protocol to compensate for the crosstalk phase error for the multiqubit simultaneous operations. $\phi_{i,0}^{(n)}$ represents the uncompensated phase for qubit Q_i. **g**, Amplitude calibration for a three-qubit simultaneous operation using the SciPy minimizer. **h**, Three-qubit SRB decays showing >99.99% primitive gate fidelity. Data are averaged over 35 randomized sequences per Clifford length. Error bars show 95% confidence intervals; fidelity uncertainties are obtained from the fitting covariance matrix.

than in the three-qubit parallel operation, with four out of five qubits achieving fidelities above 99.9%. This behaviour reflects opposing error mechanisms: at $t_g = 250$ ns, the higher total MW power required for simultaneous operation increases decoherence, probably due to heating or MW-induced noise, thereby reducing performance relative to single-qubit gates. At $t_g = 500$ ns, the lower drive amplitudes mitigate these effects. Still, the longer gate duration introduces increased dephasing from low-frequency noise, yielding fidelities comparable to single-qubit operation (see also Extended Data Table 1). The similar fidelities observed at $t_g = 250$ ns and 500 ns indicate that high-fidelity simultaneous control with ≈99.9% fidelities could be extended to ten qubits at $t_g = 500$ ns, where the total MW power matches that of five-qubit operation at $t_g = 250$ ns.

Off-resonant microwave tones can induce qubit-frequency shifts, which appear as idle-qubit phase shifts as shown in Fig. 4d. During simultaneous driven gates, however, the same frequency shifts can also detune the applied pulses, causing residual rotation errors that are not fully described by the idle-qubit phase measurements alone[37]. These residual driven-gate errors are minimized empirically by the simultaneous-drive amplitude calibration. In the five-qubit simultaneous-drive experiment, applying a simultaneous-drive amplitude calibration did not lead to a notable change in the observed fidelities, probably because lower driving amplitudes are used for these operations. In future implementations, direct compensation of the frequency shifts may provide a more systematic calibration strategy.

Conclusions

We demonstrate high-fidelity (>99.99%) single-qubit control across all five qubits in a silicon spin qubit array using tailored Kaiser-window control pulses and maintain these fidelities during simultaneous operation of up to three qubits through a shared control line. Extending to simultaneous control of all five qubits, we achieve fidelities at the practical fault-tolerant threshold of 99.9% and suggest that, based on the observed power scaling, this control approach is expected to extend to approximately ten qubits within the current device architecture. In practical quantum circuits, overall fidelity depends on the performance of all operations, including two-qubit gates, state preparation and measurement and idling. High single-qubit fidelity therefore remains beneficial for circuit performance. Together, these results demonstrate that high-fidelity parallel control across multiple silicon spin qubits

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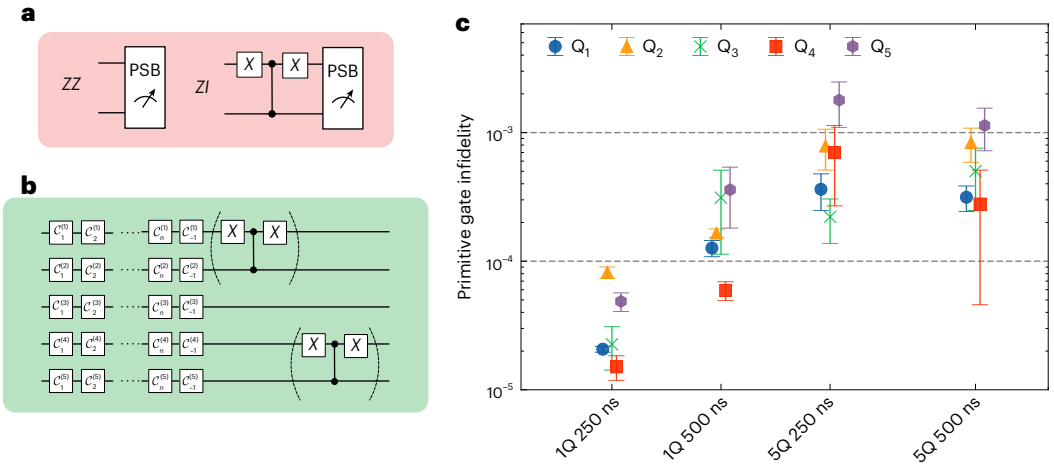


Fig. 5 | Five-qubit simultaneous operation. a, Tomographic readout protocol used to obtain the individual spin states from the PSB pairs $Q_{1,2}$ and $Q_{4,5}$. **b**, Five-qubit SRB sequence with tomographic readout. The sequence is repeated with *ZZ* and *ZI* measurements on PSB pairs $Q_{1,2}$ and $Q_{4,5}$ to reconstruct all individual qubit states. **c**, Primitive gate infidelity of single-qubit operation and five-qubit simultaneous operation with *X* gate times of 250 ns and 500 ns. Four of five qubits exceed fidelities of 99.9%, with Q_5 reaching approximately 99.9%. Points are fitted infidelities from 35 randomized sequences per Clifford length; uncertainties are obtained from the fitting covariance matrix.

is achievable, addressing a central challenge in scaling spin-based quantum processors.

This level of parallel control is enabled by a scalable calibration protocol that compensates for MW-induced, qubit-specific phase shifts from off-resonant a.c.-Stark shifts. For simultaneous control of N qubits, instead of calibrating all the $2^N - N - 1$ simultaneous driving combinations, only $N(N-1)$ pairwise calibrations are required, avoiding the exponential overhead of full crosstalk characterization. In general, the magnitude of these phase shifts scales inversely with detuning, such that corrections become negligible for sufficiently detuned qubit pairs (Extended Data Fig. 6), further reducing the calibration overhead in larger arrays. This scaling behaviour is compatible with architectures in which qubits are organized into subunits of order ten, each operated via a shared control line and driven by a common microwave source. These results thus provide guidance for scalable control architectures enabling parallel operation across multiple subunits and indicate a viable route towards scaling qubit control in larger arrays.

In a quantum processor, two-qubit operations are also required. The a.c.-Stark shift calibration method presented here applies when qubits are effectively uncoupled during microwave control, such that the resulting phase shifts correspond to additive single-qubit *Z* rotations. When exchange is activated during microwave driving, however, the phase evolution is no longer restricted to simple single-qubit phase shifts and additional conditional phases can arise[38]. Resonantly driven two-qubit gates[30] therefore involve microwave control while the qubits are coupled and are expected to exhibit additional microwave-induced conditional phases arising from a.c.-Stark effects, which would require calibration. By contrast, exchange-controlled *CZ* gates implemented with baseband pulses avoid this complication because they do not rely on microwave driving during the interaction[39].

While fidelities remain high, simultaneous five-qubit operation is currently limited by MW-induced decoherence at fast gate speeds and by dephasing at longer gate durations. Modest improvements to the micromagnet design can double the driving-field gradient (Extended Data Fig. 8 and Supplementary Section 1), enabling faster gates at reduced power. This would mitigate MW-induced decoherence and extend the dephasing-limited regime to higher speeds, allowing fidelities above 99.999% at modest gate speeds of 2–3 MHz (Fig. 2d), and supporting high-fidelity parallel control of larger qubit numbers.

To further enhance control performance and scalability, the micromagnet can be replaced by nanomagnets integrated into the gate layer or back-end-of-line wiring. Placing magnetic structures closer to the qubits provides stronger local field gradients[40,41] and preserves gradient strength in multilayer routing architectures required for 2D or quasi-2D spin qubit arrays[42,43]. Without such integration, the increased vertical separation between qubits and magnetic structures above the routing layers will otherwise degrade control performance[44,45], limiting the scalability of parallel qubit control.

Our shared control approach directly addresses a key scaling bottleneck in silicon spin qubits, enabling parallel and high-fidelity operation. Driving multiple qubits from a single impedance-matched control line reduces wiring overhead and simplifies on-chip control infrastructure. Combined with compact pairwise calibration and targeted device improvements, this offers a practical control strategy for scalable silicon qubit unit cells. Crucially, the shared-control-line architecture can be extended beyond local units to support the parallel control of multiple subarrays, with total MW drive power and calibratable a.c.-Stark shifts as the primary constraints. Together, these advances define a realistic path towards scalable, parallel qubit control in silicon, which is a core requirement for the development of large-scale quantum processors.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-026-03466-2.

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Methods

Device fabrication

The quantum dots were defined in a 9-nm-thick isotopically enriched silicon quantum well with a residual Si-29 concentration of 800 ppm. The quantum well is covered by a 30-nm SiGe buffer layer and approximately 2 nm of silicon oxide. A 15-nm Al_2O_3 layer was then deposited on the wafer using atomic layer deposition. Three layers of overlapping aluminium gates were fabricated through electron-beam lithography and lift-off processes. The aluminium was oxidized, and the resulting native aluminium oxide serves as an insulator between the gate layers. A micromagnet was fabricated on top of the device to provide magnetic field gradients for qubit addressability and operations. The micromagnet consists of a stack of 5-nm titanium and 250-nm cobalt films and is positioned on top of the overlapping gates, separated by a 30-nm atomic layer deposition aluminium oxide layer. The micromagnet design is similar to those used in our previous devices[30,33,47] with a gap size of 400 nm. See also Extended Data Fig. 8 and Supplementary Section 1 for more details.

Qubit control

To control the qubit energies, quantum dot confinements and tunnel couplings, we apply d.c. voltages using D5a digital-to-analogue converters from Qblox and use baseband pulses generated by Keysight M3201A arbitrary waveform generators to pulse the qubits between operation and readout points. Qubit control pulses are generated by Keysight M3202A arbitrary waveform generators and mixed with local oscillators from a multichannel signal generator (APMS20G4) using IQ mixers. The single-qubit operations of the five qubits are facilitated through three IQ mixers (2xMMIQ-0626HS and 1xMLIQ02181). The IQ mixers have been carefully calibrated to minimize leakage of local oscillator and image signals. The resulting signals are combined in a power splitter (ZC8PD-5R263-S+), amplified (ZVA-183G-S+) and bandpass-filtered. An MW switch (P9402C) suppresses MW leakage outside the operation stage. The MW signal is then routed through a low-loss superconducting NbTi coaxial line inside the dilution refrigerator, entering a designated impedance-controlled MW line on the sample PCB. From there, the signal is transmitted via bond wires to the quantum chip, reaching the designated MW gate (Fig. 1a).

The qubit displacement under EDSR driving can be estimated from the simulated micromagnet gradient of 1.25 mT nm^{-1}. Converting the Rabi frequency $f_R = 3$ MHz to a driving field via $B_1 = hf_R/(g\mu_B)$ with $g \approx 2$ yields $B_1 \approx 0.107$ mT, corresponding to a displacement $\Delta x \approx 0.09$ nm.

To perform the CROT and CZ two-qubit gates required for initializing, operating and reading out the quantum processor, the exchange interaction between neighbouring qubits is increased from the residual exchange $J < 16.6$ kHz (Extended Data Fig. 1) to a few megahertz using baseband pulses. In this device, spin–valley mixing occurs as the exchange interaction is increased, limiting the fidelity of our *CZ* two-qubit gates to <97%. Under single-qubit operation conditions, where only residual exchange is present, the device can be tuned to eliminate the spin–valley mixing effect across all five qubits.

Kaiser-window pulse

We use the Kaiser-window envelope, defined as

$$w(t) = \begin{cases} \dfrac{I_0\left(\beta\sqrt{1-\left(\dfrac{t-t_g/2}{t_g/2}\right)^2}\right)}{I_0(\beta)}, & 0 \le t \le t_g \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where I_0 is the zeroth-order modified Bessel function of the first kind and β is the pulse parameter. We convert from a rectangular (square) pulse of amplitude V_{rec} and gate time t_g to a Kaiser pulse of $V_{Kaiser}w(t)$ of the same gate time. The Kaiser pulse amplitude is calculated with $V_{Kaiser} = V_{rec}t_g/\mathcal{I}$, where $\mathcal{I}$ is the integral of the Kaiser window $\mathcal{I} = \int_0^{t_g} w(t)\mathrm{d}t$.

Pulse shaping also affects the accumulated a.c.-Stark phase, which scales approximately with $\int\Omega(t)^2\mathrm{d}t$, where $\Omega(t)$ is the instantaneous Rabi frequency. Consequently, pulse envelopes that produce the same qubit rotation, determined by the pulse area $\int\Omega(t)\mathrm{d}t$, can still accumulate different Stark phases depending on their temporal profiles. For a simple square pulse, the a.c.-Stark phase is expected to be minimal, but such pulses exhibit strong spectral leakage, which maximizes resonant microwave crosstalk to neighbouring qubits[26].

Randomized benchmarking

The Clifford gate set used in the RB experiments includes only *X* and *Y* gates, allowing the primitive gate fidelity to be directly attributed to the fidelity of the corresponding *X* and *Y* gates, each representing a π/2 rotation. The full Clifford set decomposition is {*XXXX*, *Y*, *X*, *YX*, *XY*, *YY*, *XX*, *YXX*, *YYX*, *YXY*, *YYY*, *XXX*, *XXY*, *XYY*, *YXYY*, *YYXY*, *XXXY*, *YXXX*, *YYXX*, *XYYY*, *YYYX*, *YYYXY*, *YXXXY*, *YXYYY*}where the identity gate is replaced with four *X* gates to prevent qubit idling. The Clifford gate set comprises 78 primitive gates when including the 4 × *X* identity replacement and 74 when using the 0 × *X* replacement. This corresponds to an average of 3.25 and 3.217 primitive gates per Clifford, respectively. In the applied sequences, the primitive gates are interleaved with 2 ns of idle time, corresponding to the minimum delay allowed by the field-programmable gate array for updating the MW phase.

We chose randomized sequences with a maximal Clifford gate length of either $2^{10} = 1{,}024$, $2^{13} = 8{,}192$, $2^{14} = 16{,}384$ or $2^{15} = 32{,}768$ for the single-qubit and the SRB experiments. For each gate length, we randomize 35 times and average over 350 single-shot measurements in each randomization to obtain the spin state probabilities. At the end of each sequence, we apply a recovery gate that ensures the entire sequence acts as either an identity operation or a X^2 flip. Each sequence then gives the corresponding flip probability $F_{noflip}(n)$ and $F_{flip}(n)$. We fit the difference $F(n) = F_{flip}(n) - F_{noflip}(n)$ with a exponential function $F(n) = Ap^n$ to obtain the depolarizing parameter p (refs. 48,49). The fitted parameter A is used to normalize the raw data for plotting. We calculate the Clifford fidelity with $F_C = (1 + p)/2$ and obtain the primitive gate fidelity by $F_p = 1 - (1 - F_C)/3.25$ (ref. 13). To ensure the validity of fitting with the Ap^n model, we also perform fits using an additional offset parameter, $Ap^n + B$. For all qubits, the fitted offsets satisfy $|B| < 0.015$. The difference in the resulting primitive gate fidelities is limited to one decimal place smaller than our reported significant figures, thereby validating the use of the Ap^n fitting formula.

Within a single randomization, we load all sequences of varying lengths. This results in a total of approximately two million primitive gates per loaded waveform. The loading process takes roughly 5 min, and a full RB measurement requires about 3 h. No calibration of the qubit resonance frequency is performed during this 3-h period, indicating highly stable qubit conditions. The number of randomizations $K = 35$ is determined to provide an RB accuracy of $\mu = \log_{10}(\hat{r}/r) \approx 0.1$, where $\hat{r}$ is the RB-estimated average error rate and r is the theoretical average error rate when given $r = 10^{-4}$ (ref. 50). This number of randomizations is sufficient to resolve the 10^{-4} error rate reported in this work. The precise measurement of a 10^{-5} error rate would require sequences with longer gate length and additional data points along the decay curve.

SRB is performed by implementing single-qubit RB experiments in parallel on multiple qubits. For each qubit, distinct RB sequences are generated, each composed of π/2 gates with identical gate times. These sequences are temporally synchronized, enabling a synchronized operation cycle with simultaneous operations across multiple qubits. Due to the generation of different RB sequences, the number of primitive gates in each sequence may vary across qubits. Any additional primitive gates are applied either through simultaneous pulses on a subset of qubits or via individual single-qubit pulses. The difference in the number of primitive gates among individual qubit sequences can be as large as 200 for a sequence containing 10,000 Clifford gates.

Each qubit's sequence is analysed independently, similar to a single-qubit RB experiment, to determine the fidelity of individual single-qubit primitive gates. This separated analysis helps identify qubit-specific errors. A joint fidelity is obtained by multiplying the individual fidelities.

Five-qubit tomographic readout

The five-qubit operation cycle provides only 3 out of 5 bits of information (2 PSB and 1 QND readout). Therefore, a tomographic readout is required to fully measure the five-qubit spin state in the five-qubit SRB experiment. To achieve this, we repeat each operation sequence twice: one *ZZ* component read out and one *ZI* component read out. The *ZZ* component is measured directly with the *ZZ* measurement operator of the PSB readout. The *ZI* component is obtained by mapping the *ZZ* operator to a *ZI* operator by applying a CNOT gate before the PSB readout. The two-qubit state probabilities of qubit pairs $Q_{1,2}$ and $Q_{4,5}$ are used to construct single-qubit probabilities $P_{1,\uparrow}$, $P_{2,\uparrow}$, $P_{4,\uparrow}$ and $P_{5,\uparrow}$. We synthesize the CNOT gate using a *CZ* gate and single-qubit gates[12].

By thresholding the two single-shot measurements s_{ZZ} and s_{ZI} with the readout threshold value v_{thres}, we obtain the two-qubit state probabilities

$$P_{\uparrow\uparrow} = (s_{ZZ} > v_{\mathrm{thres}})\&(s_{ZI} > v_{\mathrm{thres}}), \tag{2}$$

$$P_{\downarrow\uparrow} = (s_{ZZ} < v_{\mathrm{thres}})\&(s_{ZI} < v_{\mathrm{thres}}), \tag{3}$$

$$P_{\uparrow\downarrow} = (s_{ZZ} < v_{\mathrm{thres}})\&(s_{ZI} > v_{\mathrm{thres}}), \tag{4}$$

$$P_{\downarrow\downarrow} = (s_{ZZ} > v_{\mathrm{thres}})\&(s_{ZI} < v_{\mathrm{thres}}). \tag{5}$$

We then calculate the single-qubit state probability by

$$P_{1,\uparrow} = P_{\uparrow\downarrow} + P_{\uparrow\uparrow}, \tag{6}$$

$$P_{2,\uparrow} = P_{\downarrow\uparrow} + P_{\uparrow\uparrow}. \tag{7}$$

Similarly, we obtain the probabilities $P_{4,\uparrow}$ and $P_{5,\uparrow}$. The sequence fidelity, which is fitted in the five-qubit SRB experiments, is calculated by $F_{\mathrm{seq},i} = |P_{i,\uparrow}^{\mathrm{flip}} - P_{i,\uparrow}^{\mathrm{noflip}}|$.

RB simulations incorporating measured qubit noise

To model the impact of noise on the RB fidelities, we first extract a noise power spectral density (PSD) model from Ramsey and Carr–Purcell–Meiboom–Gill measurements performed under idling conditions, during which the quantum processor was not actively operated (Extended Data Fig. 4a). Based on this PSD, we generate noise contributions across three characteristic timescales determined by the experimental conditions: low-frequency (LF) noise (Extended Data Fig. 4b), intermediate-frequency (IF) noise and high-frequency (HF) noise.

RB sequences are constructed using the same gate set as in the experiments. Each primitive gate is simulated as a unitary time evolution $\exp(-iH_{\mathrm{op}}(t)t_g)$, where $H_{\mathrm{op}}(t)$ includes both the ideal drive and a σ_z detuning[51] induced by noise:

$$H_{\mathrm{op}}(t) = 2\pi\left(\frac{\beta(t)}{2}\sigma_z + \frac{f_{\mathrm{R}}}{2}(\cos(\phi)\,\sigma_x + \sin(\phi)\,\sigma_y)\right),$$

with $\beta(t) = \beta_{\mathrm{LF}} + \beta_{\mathrm{IF}} + \beta_{\mathrm{HF}}(t)$.

Here, $\beta(t)$ describes the detuning noise in hertz. The noise is incorporated into the simulation as follows: the LF noise component β_{LF} is held constant over each RB sequence and updated between sequences, while the IF component β_{IF} is updated after each primitive gate. To realistically capture correlations across timescales, LF and IF noise are generated jointly rather than independently. HF noise $\beta_{\mathrm{HF}}(t)$ is modelled as a fluctuating detuning during gate execution, updated every 10-ns or 50-ns time step. Because HF noise has no measurable impact on the simulated fidelities at the experimentally relevant levels, it is neglected in the simulations.

After simulating the complete qubit evolution for an RB sequence under noise, we calculate the return probabilities F_{noflip} and F_{flip} and extract RB fidelities using the same procedure as for the experimental data.

The observed dependence of gate fidelity on gate duration can be understood analytically by considering the effect of slow detuning noise in the quasi-static regime. In this limit, the frequency detuning β remains effectively constant during a gate of duration t_g, leading to a relative phase accumulation $\Delta\phi = 2\pi\beta t_g$ between the computational basis states[52]. For small $\Delta\phi$, and assuming a symmetric distribution of detunings centred at zero, the π/2-gate infidelity becomes $1 - F = 1 - |\langle\psi_{\mathrm{ideal}}|\psi\rangle|^2 \propto \langle(\Delta\phi)^2\rangle \propto \langle\beta^2\rangle t_g^2$, linking gate error directly to the noise variance $\langle\beta^2\rangle$ and the square of the gate duration[53]. This quadratic dependence on t_g is observed both in simulation and in experimental data where dephasing dominates (t_g = 250 and 500 ns) (Fig. 2d).

Data availability

The data that support the findings of this study are openly available via Zenodo at https://doi.org/10.5281/zenodo.21507661 (ref. 54). Source data are provided with this paper.

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Acknowledgements

We thank M. Rimbach-Russ and L. M. K. Vandersypen for helpful discussions, and J. S. Rojas-Arias for assistance with the RB model. We also acknowledge R. Kuroda for support with sample fabrication.

Author contributions

Y.H.W., L.C.C. and P.B. performed the experiment. A.N. fabricated the device. I.K.J., K.T., A.N., T.K., T.N. and H.-S.G. contributed to the data acquisition and discussed the results. G.S. developed and supplied the silicon-28/silicon-germanium heterostructure. Y.-H.W. and L.C.C. wrote the paper with inputs from all co-authors. S.T. supervised the project.

Funding
This work was supported by JST Moonshot R&D grant number JPMJMS226B. L.C.C. acknowledges support from the Swiss National Science Foundation Mobility Fellowship number P2BSP2_200127, and Y.H.W. acknowledges support from RIKEN's IPA programme. We also acknowledge support from JSPS KAKENHI grant numbers 22H01160 (T.K.) and 23H05455 (K.T.), JST PRESTO grant numbers JPMJPR2017 (T.N.) and JPMJPR23F8 (A.N.). H.-S. Goan acknowledges support from the National Science and Technology Council, Taiwan, under grant numbers NSTC 113-2112-M-002-022-MY3, NSTC 113-2119-M-002-021, 114-2119-M-002-018, NSTC 114-2119-M-002-017-MY3 and NSTC 115-2119-M-002-005 and from the National Taiwan University under grant numbers NTU-CC-115L8937, NTU-CC-115L893704 and NTU-CC-115L8512. H.-S.G. is also grateful for the support of the 'Center for Advanced Computing and Imaging in Biomedicine (NTU-114L900702 and NTU-115L900702)' through the Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE), Taiwan, the support of Taiwan Semiconductor Research Institute (TSRI) through the Joint Developed Project (JDP) and the support from the Physics Division, National Center for Theoretical Sciences, Taiwan.

Competing interests
The authors declare no competing interests.

Additional information
Extended data is available for this paper at https://doi.org/10.1038/s41567-026-03466-2.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41567-026-03466-2.

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Peer review information *Nature Physics* thanks Ross Leon and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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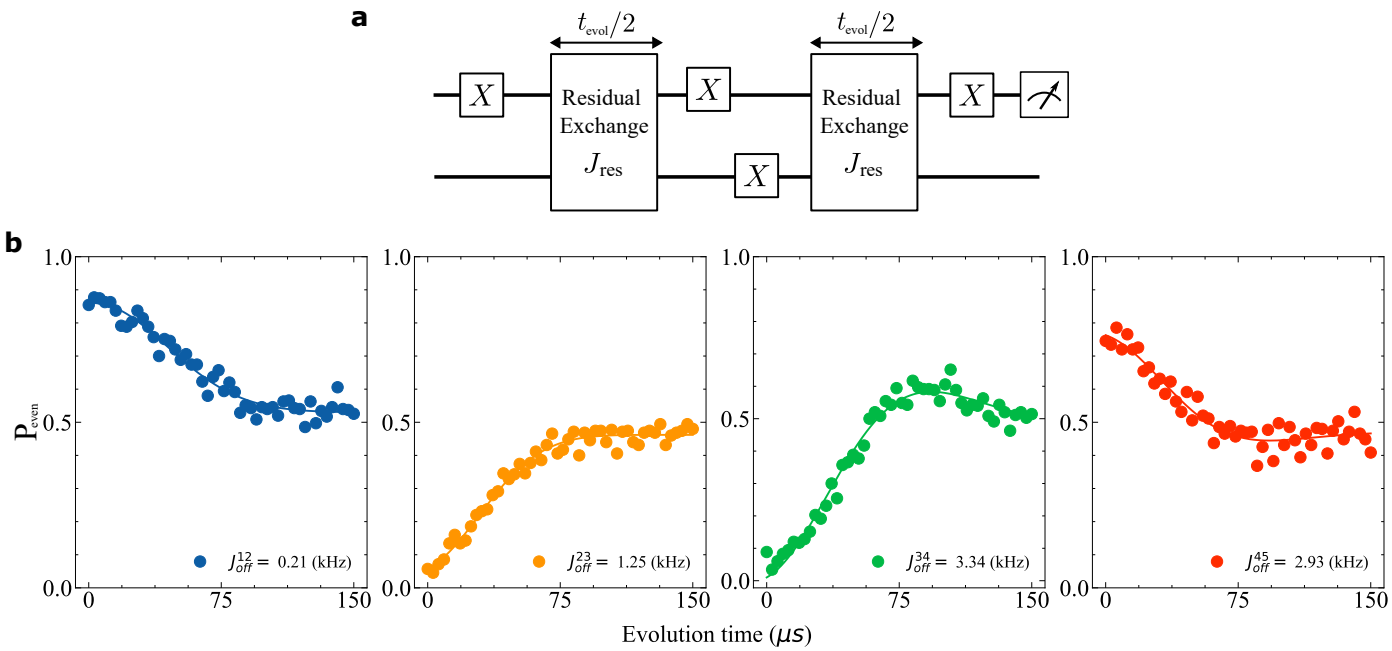
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Extended Data Table 1 | Overview of measured primitive gate fidelities

Driving condition	Q1	Q2	Q3	Q4	Q5
1-qubit (83 ns)	99.96(2)	99.9979(2)	99.9982(2)	99.9957(4)	99.982(7)
1-qubit (125 ns)	99.993(1)	99.9955(4)	99.9987(4)	99.9984(3)	99.9943(7)
1-qubit (250 ns)	99.9979(1)	99.9918(8)	99.9977(8)	99.9985(3)	99.9943(7)
1-qubit (500 ns)	99.987(2)	99.984(1)	99.97(2)	99.994(1)	99.96(2)
2-qubit (Q2,3, 125 ns)	N/A	99.9916(4)	99.9930(2)	N/A	N/A
2-qubit (Q3,4, 125 ns)	N/A	N/A	99.991(3)	99.9970(4)	N/A
2-qubit (Q1,3, 250 ns)	99.9937(6)	N/A	99.991(1)	N/A	N/A
2-qubit (Q3,5, 250 ns)	N/A	N/A	99.9950(11)	N/A	99.9911(9)
3-qubit (Q2,3,4, 250 ns)	N/A	99.9926(7)	99.9929(16)	99.9966(7)	N/A
3-qubit (Q1,3,5, 250 ns)	99.992(1)	N/A	99.995(2)	N/A	99.983(2)
5-qubit (250 ns)	99.96(1)	99.92(3)	99.977(8)	99.93(4)	99.82(7)
5-qubit (500 ns)	99.978(4)	99.91(2)	99.96(2)	99.94(5)	99.8(1)

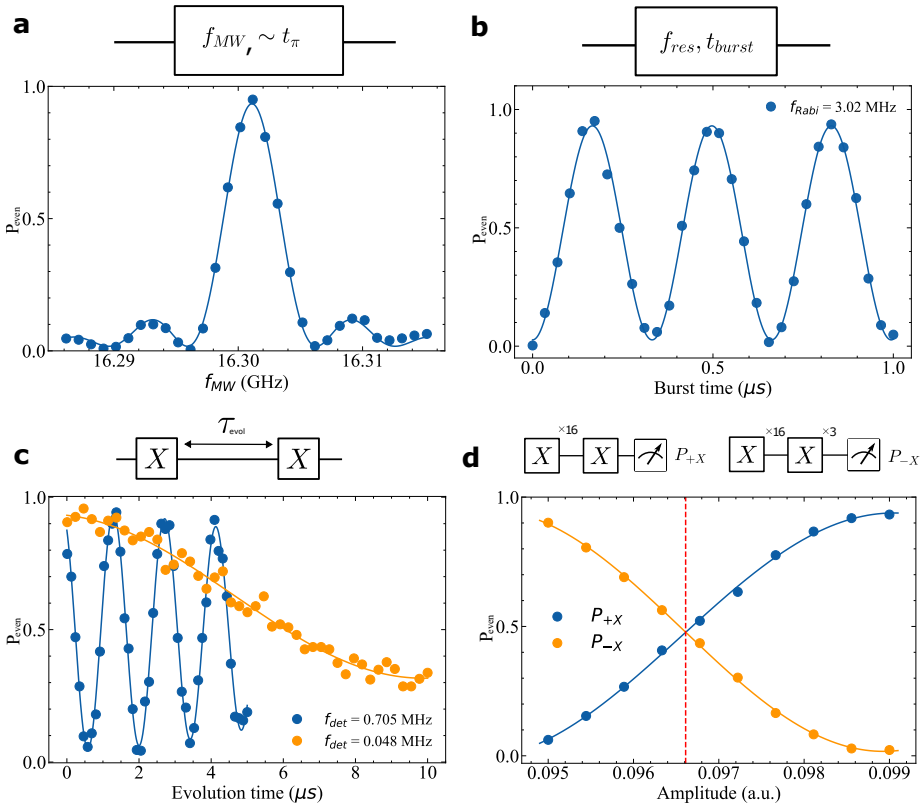
Primitive randomized benchmarking fidelities for all five qubits under various driving conditions, including isolated (1-qubit), pairwise (2-qubit), three-qubit (3-qubit), and fully parallel five-qubit operation (5-qubit). Control fidelities are shown for pulse durations of 83-500 ns used in these experiments.

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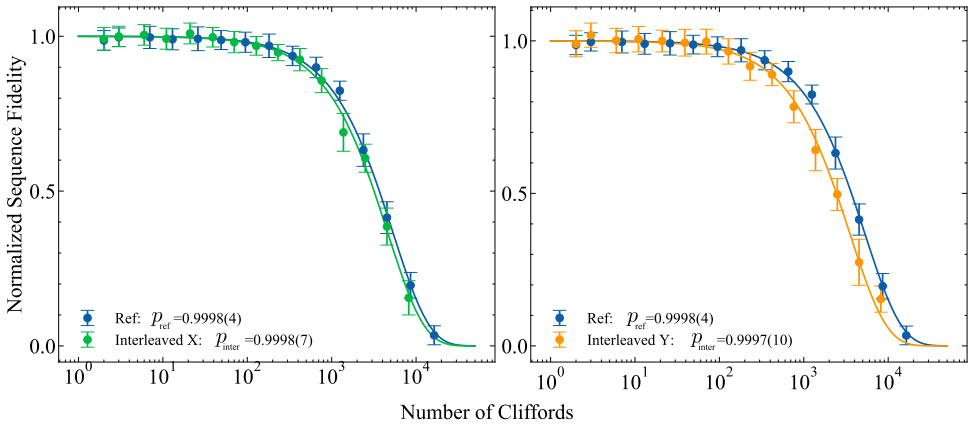
Extended Data Fig. 1 | Residual exchange of the qubit pairs. a Decoupled CZ (DCZ) sequence used to measure residual exchange J_{off}, defined as the unintended exchange interaction between qubits during single-qubit operations. This protocol isolates the effect of residual coupling by removing dynamic phase accumulations. **b** Residual exchange measured in the four qubit pairs. In the presence of exchange, coherent oscillations would be expected. However, all measurements show no notable oscillations over the echo coherence time of ≈30 μs. The observed decay is consistent with Hahn echo dephasing, setting an upper bound on the residual exchange of $J_{off} < (1/30\ \mu s)/2 \simeq 16.6$kHz.

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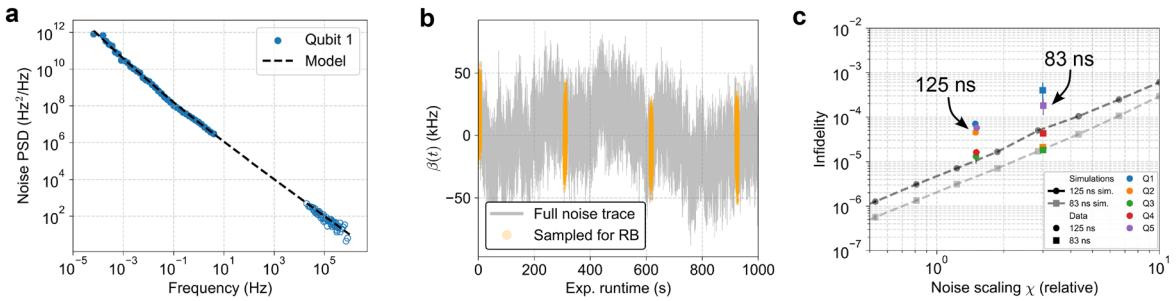
Extended Data Fig. 2 | Single-qubit gate calibration. a The resonance frequency is roughly tuned by applying a MW burst of gate time $t_g \approx t_\pi$ with varied frequency f_{MW} and measure the returned even parity probability P_{even}. This probability is then fitted with equation $P_{even}(f_{MW}) = A(f_R^2/((f_{res} - f_{MW})^2 + f_R^2)\sin^2(\pi t_{pulse}\sqrt{((f_{res} - f_{MW})^2 + f_R^2)})) + B$ to obtain resonance frequency f_{res}. **b** The pulse amplitude is roughly tuned by measuring the Rabi oscillation with an amplitude A and fit with $P_{even}(t_{burst}) = A\cos(2\pi f_R t_{burst}) + C$ to get the Rabi frequency f_R. The pulse amplitude is adjusted to $A' = A \times (f_R^{target}/f_R)$ for a chosen target Rabi frequency f_R^{target}. The $\pi/2$ pulse gate time is set to $t_g = 0.25/f_R^{target}$. **c** The frequency of the pulse is fine-tuned with a Ramsey sequence. We intentionally offset the MW frequency with a detuning of $\Delta f = 0.7$ MHz and apply the Ramsey sequence with pulse time t_g. The even parity probability P_{even} of the Ramsey measurement oscillates with a frequency equal to the detuning from the resonance. We fit this oscillation with $P_{even}(t_{evol}) = A\cos(2\pi f_{det} t_{evol} + \phi)\exp(-(t/T_2^*)^2) + C$ and obtain the detuning f_{det}. We then shift the MW frequency towards the qubit resonance frequency by the amount of $\Delta f - f_{det}$. After this fine-tuning of the MW frequency, the Ramsey measurement shows a reduced detuning of $f_{det} = 48$ kHz. **d** To fine-tune the pulse amplitude, we apply a calibration sequence of $16\pi/2$ pulses, followed by either one or three additional $\pi/2$ pulses to rotate the spin into the $|+x\rangle$ or $|-x\rangle$ state, respectively. We then vary the pulse amplitude V_{amp} and fit the resulting return probabilities with a sine function $P_{even}(V_{amp}) = A\cos(kV_{amp} + \phi) + B$. The optimal amplitude is determined from the intersection point of the two fitted curves.

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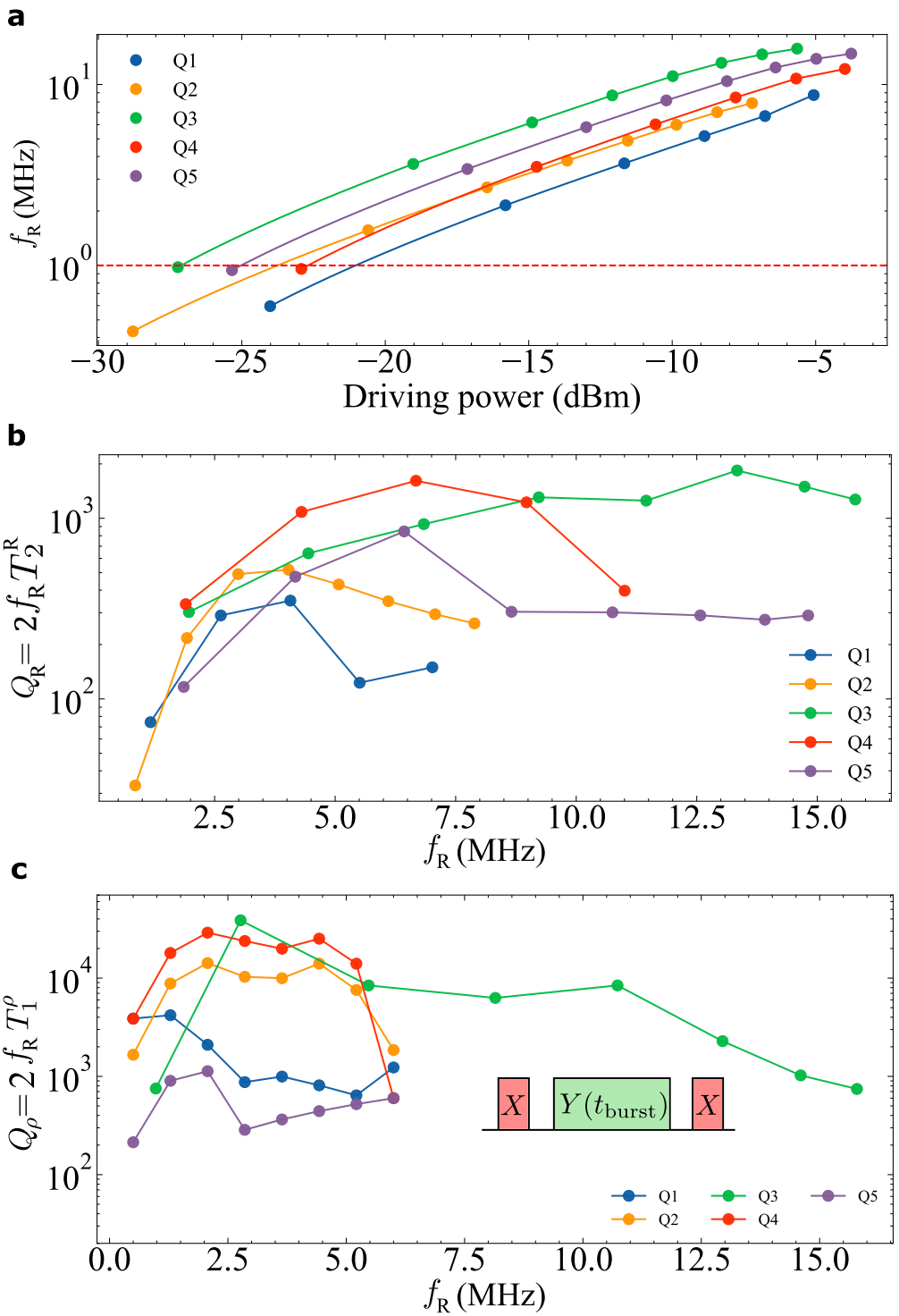
Extended Data Fig. 3 | Single-qubit interleaved RB. Measurements of the individual fidelity of the primitive $X(\pi/2)$ and $Y(\pi/2)$ single-qubit gates using interleaved randomized benchmarking. Because the X and Y gates differ only by a phase shift implemented in the AWG sequences that generate the MW signals using IQ mixers (Methods), they are expected to exhibit identical performance. The interleaved gate fidelity is calculated from the fitted depolarizing parameters of the reference p_{ref} and interleaved p_{inter} randomized benchmarking sequences using $F_{inter} = ((d-1)(1+p_{inter}/p_{ref}))/d$. The fitted depolarizing parameters are $p_{ref} = 0.9998(4)$, $p^{X}_{inter} = 0.9998(7)$ and $p^{Y}_{inter} = 0.9997(10)$. The resulting primitive gate fidelities are 99.9985(8)% for the X gate and 99.995(1)% for the Y gate. Data are averaged over 35 randomized sequences per Clifford length. Error bars show 95% confidence intervals. Uncertainties in the fitted depolarizing parameters are obtained from the fitting covariance matrix, and uncertainties in the interleaved gate fidelities are calculated by error propagation.

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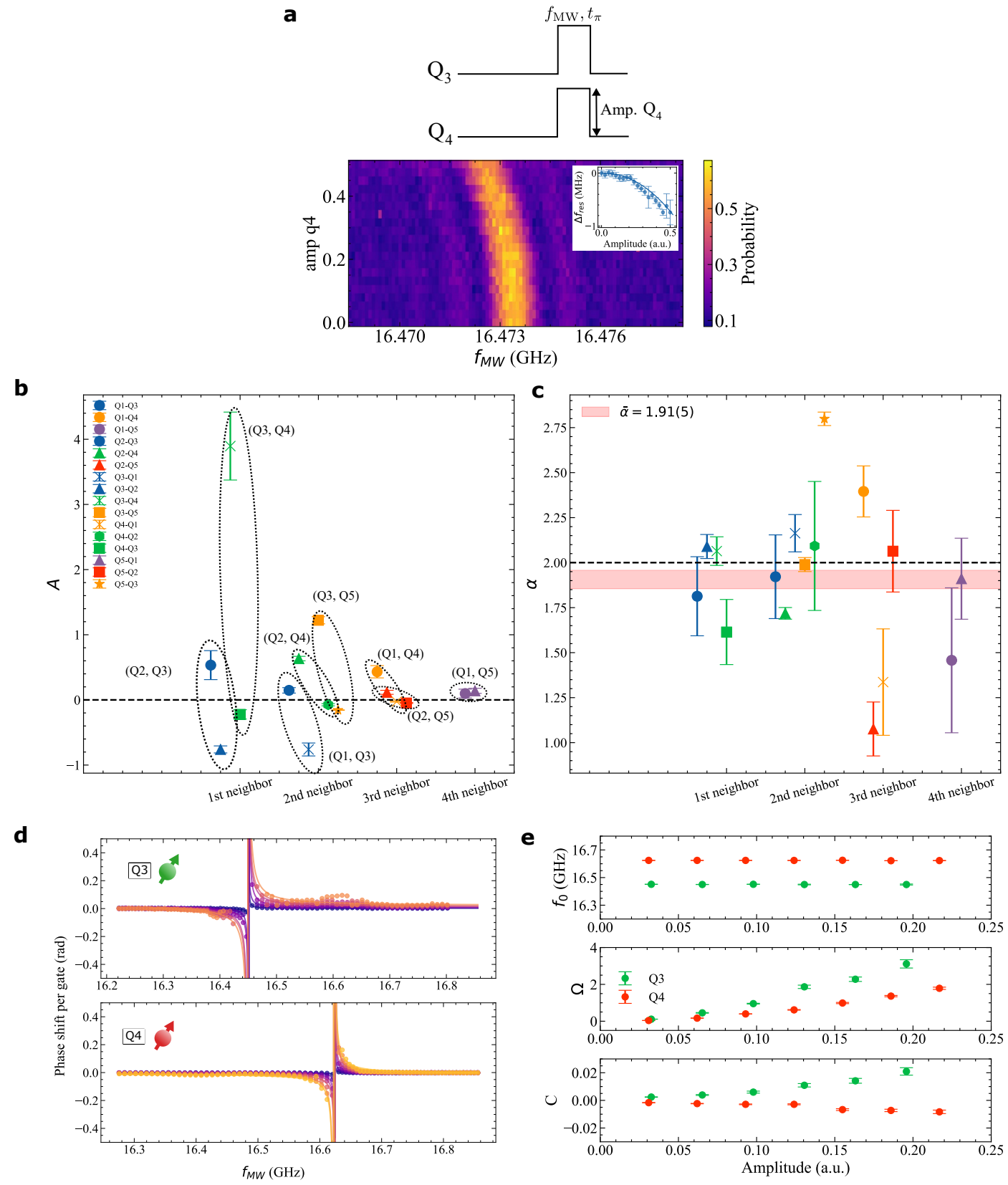
Extended Data Fig. 4 | Simulations and noise-limited primitive gate fidelity. **a** Power spectral density (PSD) of qubit Q1, extracted from Ramsey (filled circles, low-frequency) and CPMG (open circles, high frequency) measurements [20,31], plotted as the positive-frequency part of the two-sided spectrum. Other qubits exhibit similar noise characteristics. A combined fit using $S(f) = 10^7(0.25 \cdot f^{-1.3} + f^{-1})$ accurately reproduces the measured spectrum and is used to generate the simulated time-domain traces in (b). Notably, this PSD reflects noise under idling conditions, when the processor is not actively operated. **b** Simulated low-frequency noise trace $\beta(t)$ generated from the PSD in (a), using the same parameters as in the experimental randomized benchmarking (RB) measurements: 500 single-shot measurements, 20 randomized sequences, 10 gate lengths, and a 1.75 ms RB cycle time for both recovery-gate sequences projecting into the up and down states. Long AWG loading times of approximately 5 minutes are also included. Orange traces indicate the subset of the noise used for RB simulations; only four repetitions are shown for clarity. **c** Simulated primitive $\pi/2$ fidelities for different noise strengths. The noise scaling factor χ rescales the PSD as $S_\chi(f) = \chi^2 S_{\chi=1}(f)$, corresponding to a linear scaling of the standard deviation. For gate times of 125 ns (83 ns), reproducing the experimental fidelities requires scaling the noise standard deviation by roughly 1.5 × (3 ×). For longer gate times, the noise model captures the experimental fidelities rather well (see Fig. 2(d)).

https://doi.org/10.1038/s41567-026-03466-2



Extended Data Fig. 5 | Qubit coherence when driven. a Qubit Rabi frequency power dependence. The driving amplitude V in Fig. 1(h) is converted into power P by the relation $P(V) = 10\log_{10}((V/\alpha)^2/50)$, where α is the qubit-specific impedance scaling factor. This factor is derived by matching the observed voltage V_1MHz to the power P_1MHz (referenced to 50Ω) measured by a spectrum analyzer: $\alpha = V_{1\mathrm{MHz}}/\sqrt{(50 \cdot 10^{(P_1\mathrm{MHz}/10)})}$. **b** Rabi oscillation quality factors $Q_R = 2f_R T_2^R$ under different driving Rabi frequency f_R. The Rabi coherence times are obtained by fitting the decays of Rabi oscillation amplitudes with $\exp(-t/T_2^R)W(t)$ where $W(t) = (1 + t^2/(f_R(T_2^*)^2)^2)^{-1/4}$ takes into account the effect of dephasing [32]. **c** Spin-lock quality factor $Q_\rho = 2f_R T_1^\rho$. The spin-lock decay time T_1^ρ is measured by first flipping the spin to the y-axis with an X gate, followed by a drive in Y-direction of duration t_{burst} and another X gate projecting the spin back to the principal axis.

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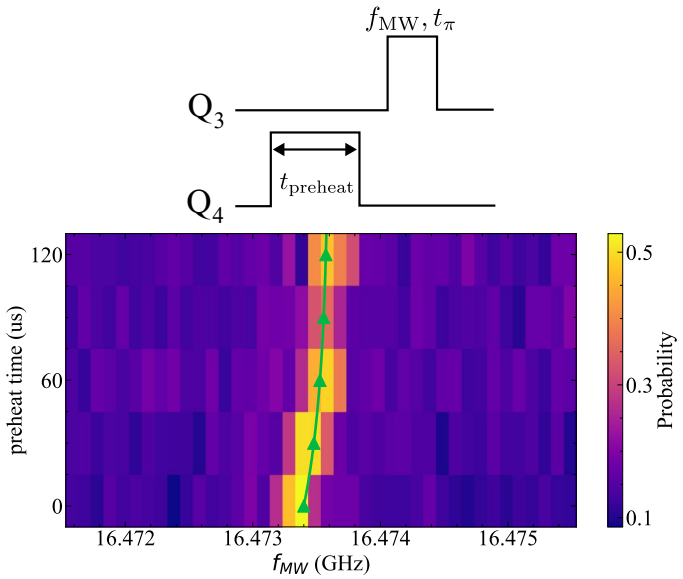


Extended Data Fig. 6 | See next page for caption.

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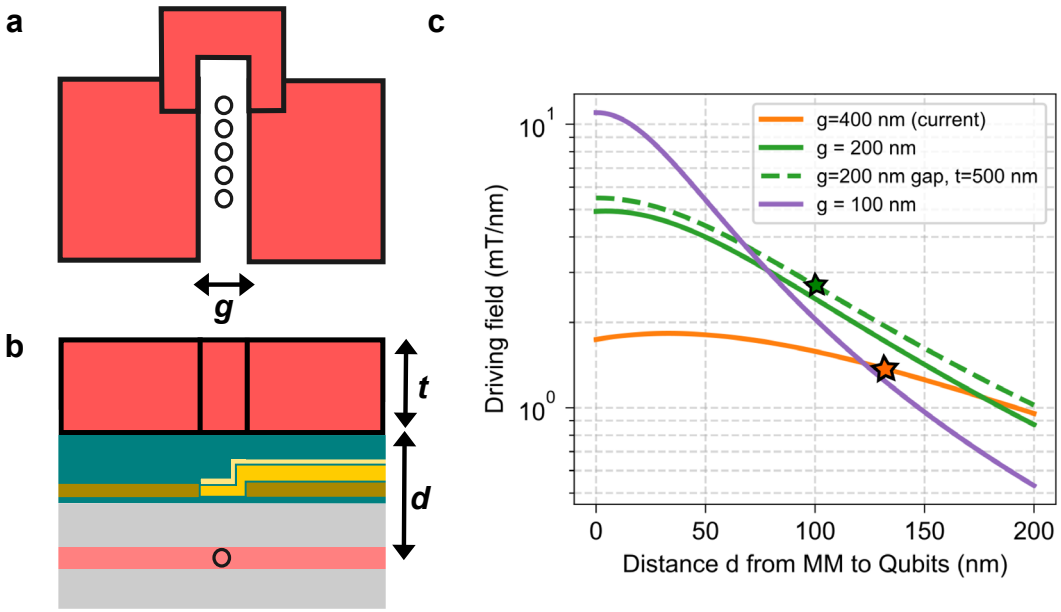
Extended Data Fig. 6 | AC-Stark shift. a Resonance frequency shift of Qubit Q_3 when driven simultaneously with an X^2 gate on Q_4 (166 ns duration). The shift of ≈− 0.8 MHz exceeds the ≈− 0.2 MHz shifts caused by heating due to MW control pulses (see Extended Data Fig. 7), indicating a distinct mechanism. Inset: Fit to the resonance frequency shift relative to f_{MW} = 16473.5 MHz. The frequency shift is fitted with A V_{amp}^{α} resulting in $A = -3.2(3)$ and $\alpha = 2.15(5)$. **b**, **c** Amplitude dependence of crosstalk-induced phase shifts between qubit pairs, fit with A V_{amp}^{α}. The label $Q_i - Q_j$ indicates the phase shift on Q_i when driving Q_j. The extracted average exponent $\alpha = 1.91(5)$ is consistent with the expected quadratic scaling of the AC-Stark effect. As anticipated, the sign of the phase shift depends on detuning: it is positive when Q_j is higher in frequency than Q_i ($i < j$), and negative when lower ($i > j$), see panel (d). The large variation in the fitted α values is likely related to the small phase shifts measured for qubit pairs with large frequency separation. Error bars are calculated from the fitting covariance matrix. **d** Crosstalk phase shift introduced by second-tone driving with fixed amplitude and varied frequency. As the second-tone frequency is swept across the qubit resonance, the induced phase shift changes sign. The observed phase shifts caused by the AC-Stark effect follow the form $\Omega^2/(x - f_0) + C$, where Ω is the driving strength, f_0 is the resonance frequency, and C is a phenomenological offset; a positive (negative) detuning results in a positive (negative) phase shift. **e** Fitted parameters from (d). Error bars are obtained from the fitting covariance matrix.

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Extended Data Fig. 7 | MW-Heating induced frequency shift. Microwave-induced frequency shifts are commonly observed in Si/SiGe spin qubit experiments and can substantially impact the performance of quantum processors. Here, we show a resonance frequency shift of idling Qubit Q_3 when a pre-heating pulse is applied to Q_4 at a drive strength of $f_R = 3$ MHz. A frequency shift of approximately 0.22 MHz is observed for the longest heating pulse duration of $t_{\mathrm{preheat}} = 120\ \mu$s. This measurement was performed at an elevated mixing chamber temperature of approximately 120 mK, consistent with the conditions used in the RB experiments presented in this work. At this temperature, MW-heating-induced frequency shifts have been reported to be reduced [36]. Green curve corresponds to an exponential model of $f_0 + \Delta f_{\mathrm{max}}(1 - e^{(-t_{\mathrm{preheat}}/\tau)})$ with $f_0 = 16473.4$ MHz, $\Delta f_{\mathrm{max}} = 200$ kHz and $\tau = 60\ \mu$s. We observe that these shifts scale with gate speed due to increased MW power, but remain substantially smaller than those reported in other experiments.

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Extended Data Fig. 8 | Micromagnet design and simulated improvements in driving field gradient strength. a Top-down schematic of the micromagnet (MM) structure used for local spin control, with the micromagnet gap g indicated [47]. Qubit positions are illustrated as circles. This illustration is not drawn to scale. **b** Cross-section schematic of the device gate stack, showing three overlapping aluminum gate layers (shaded from brown to yellow) and the underlying 9 nm quantum well (red), where the qubits are located. The well is separated from the gates by a 30 nm SiGe spacer (gray). Oxides (pine-colored) electrically isolate the gate stack and the micromagnet. The MM thickness t and qubit-MM separation d are indicated for reference. This illustration is not to scale. **c** Simulated slanting field gradients for different micromagnet gap sizes and thicknesses. The current device (400 nm gap, 250 nm thickness) achieves a slanting field of approximately 1.4 mT/nm at a vertical distance that spatially varies between 100-160 nm (orange star at 130 nm), depending on the number of underlying metal gate layers. For simplicity, we assume a flat surface in the simulations, corresponding to a uniform separation between the qubit and micromagnet. By reducing the qubit-MM separation to 100 nm, through a thinner gate stack, and using a 200 nm gap, our simulations predict a potential doubling of the slanting field gradient. Further enhancement is possible by increasing the MM thickness, as demonstrated for the 200 nm gap and 500 nm thickness case. Stronger gradients enable faster gate operations at lower MW drive power.